

MAT 426: Topics in Linear Algebra

Exam 2

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Page:	1	2	3	4	5	6	7	Total
Points:	16	19	12	11	12	18	12	100
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- *You are NOT ALLOWED to use your book, notes and homework*
- *You are NOT ALLOWED use internet/web resources.*
- *You are NOT ALLOWED to consult with others.*
- *You are allowed to use a scientific, non graphing calculator.*

Honesty Pledge

I affirm that the work I am submitting upholds the Centre College standard for academic honesty, that it is solely my own work, and that I have neither given nor received any unauthorized assistance from other people or resources.

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1. (6 points) Answer the following by circling either TRUE or FALSE.

- | | | |
|------|-------|--|
| True | False | Every matrix has an orthogonal diagonalization. |
| True | False | Gram-Schmidt orthogonalization takes a basis for a subspace and produces an orthogonal basis for the same subspace. |
| True | False | Gram-Schmidt and QR via Householder matrices always produce an orthogonal matrix Q of the same size. |
| True | False | When implemented in a computer, the vector $\mathbf{x}_i$ used to create zeros in column i using Householder matrices is typically normalized by dividing by the largest entry in $\mathbf{x}_i$. |
| True | False | The eigenvalues of a Householder matrix H are ± 1 . |
| True | False | The singular values of a matrix A can be negative. |

2. (10 points) Let $W = \text{Span} \left\{ \mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \mathbf{u}_3 = \begin{bmatrix} 0 \\ -1 \\ 1 \\ -1 \end{bmatrix} \right\}$. Write $\mathbf{y} = \begin{bmatrix} 3 \\ 4 \\ 5 \\ 4 \end{bmatrix}$ as the sum of a vector in W and a vector orthogonal to W .

3. (10 points) Find the matrix Q in the QR decomposition of the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$.

4. (a) (5 points) How is the QR factorization of a matrix A used to determine the eigenvalues of A ?

- (b) (4 points) Without providing proof, what does the iteration described in (a) converge to?

5. (a) (2 points) Let $\mathbf{u}$ be a nonzero vector in $\mathbb{R}^n$. Define a Householder matrix H .

(b) (10 points) Let $A = \begin{bmatrix} 2 & -1 & -2 \\ -4 & 6 & 3 \\ -4 & -2 & 8 \end{bmatrix}$. Complete the following steps that use Householder matrices to find the matrix R in the QR factorization of A (you do not have to find the matrix Q or R - just complete the requested steps).

- Multiply A by a Householder matrix H_1 to obtain $A^{(1)} = \begin{bmatrix} -6 & 3 & 8 \\ 0 & 4 & -2 \\ 0 & -4 & 3 \end{bmatrix}$.

- What is the vector $\mathbf{u}_2$ that would be used to zero out the entry in row 3, column 2 of $A^{(1)}$?
- Without explicitly forming the required matrices, how would the matrix Q be determined? You do not have to compute Q .

6. (a) (3 points) Define what it means for an $n \times n$ matrix A to be orthogonally diagonalizable.
- (b) (8 points) Suppose A is symmetric. Let $\lambda_1 \neq \lambda_2$ be two eigenvalues of A and suppose that $\mathbf{v}_1$ is in the eigenspace for λ_1 and $\mathbf{v}_2$ is in the eigenspace for λ_2 . Prove that $\mathbf{v}_1$ and $\mathbf{v}_2$ are orthogonal.

7. (12 points) Let $A = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$. The characteristic equation of A is $c_A(\lambda) = -(\lambda - 7)^2(\lambda + 2)$. The eigenspace for $\lambda = -2$ and is given by

$$E_{-2} = \text{Span} \left\{ \begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix} \right\}.$$

Find the orthogonal diagonalization of A . Clearly state the matrices Q and D .

8. (a) (4 points) Identify the matrix A associated with the quadratic form $Q(\mathbf{x}) = x^2 + 7y^2 - 3z^2 + xy - 2xz + 3yz$.
- (b) (6 points) Consider the quadratic form $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ where $A = \begin{bmatrix} 1 & -4 \\ -4 & -5 \end{bmatrix}$. Classify $Q(\mathbf{x})$ as one of positive semidefinite, negative semidefinite or indefinite.
9. (8 points) Let A be an $n \times n$ symmetric matrix. Prove that there exists an orthogonal change of variables $\mathbf{x} = P\mathbf{y}$ that transforms the quadratic form $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ into a quadratic form with no cross-product terms.

10. (12 points) Let $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & -3 & 0 \\ 1 & 0 & 1 \end{bmatrix}$. The eigenvalues of A are $\lambda = 0, 4, 9$ with corresponding eigenvectors

$$\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

respectively. Find the singular value decomposition of A . Be sure to clearly state U , V , and Σ .